

Edge-Based Defense in Networks

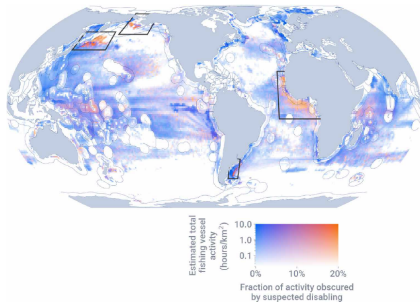
A Modification of Bloch, Chatterjee & Dutta (2023)
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Strategic adversaries on networks



Estimated global fishing activity.

Orange bands mark suspected illegal activity — West Africa is one of three concern zones.

Illegal fishing, logging, trafficking.

Shared resources reached through corridors that several authorities watch at once.

The adversary is strategic.

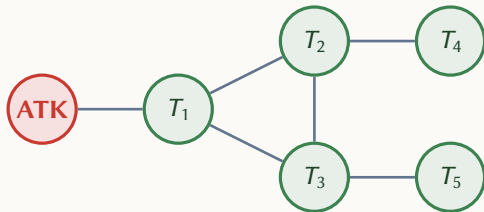
Routes are chosen deliberately. Defense is weakest where the attack will go.

Defense is shared.

Each corridor is monitored from both sides.

What is a network?

A network is a graph $G = (V, E)$: nodes V are locations, edges E are direct connections between them.



Nodes.

Locations the attacker can reach.

Edges.

The connections between locations.

The attacker.

A single adversary outside the network.

The defenders.

One at each target. They invest to intercept.

Two traditions, one gap

Node Defense		Edge Defense
Centralized	Continuous	Centralized
Dziubiński & Goyal Cerdeiro et al.	Bloch, Chatterjee & Dutta (2023)	Washburn & Wood Mavronicolas et al.
Binary, all-or-nothing. Single planner.	Continuous investment. Many defenders.	Discrete or budgeted. Single planner.

The gap — *Continuous, decentralized investment — placed on the connections.* 4 / 5

The Bloch baseline

$$\alpha_{k(p)}(p) = \prod_{\substack{j \in V(p) \\ j \neq 0, j \neq k(p)}} (1 - x_j)$$

$$U(\pi, x) = \sum_{p \in P} \pi(p) \beta_{k(p)}(p) b_{k(p)}$$

$$x_i = 1 - \frac{U}{b_i}, \quad q_i = \frac{1}{b_i} \left(1 - \frac{U}{b_i} \right)$$

Path survival.

The attacker survives the interior of the path if it slips past every defender between node 0 and the target.

Attacker's expected payoff.

Sum over every path, weighted by how often the attacker uses it, times the chance of getting through, times the target's value.

Equilibrium for a first target.

U is the attacker's payoff. Higher-value targets invest more and are attacked more often.